



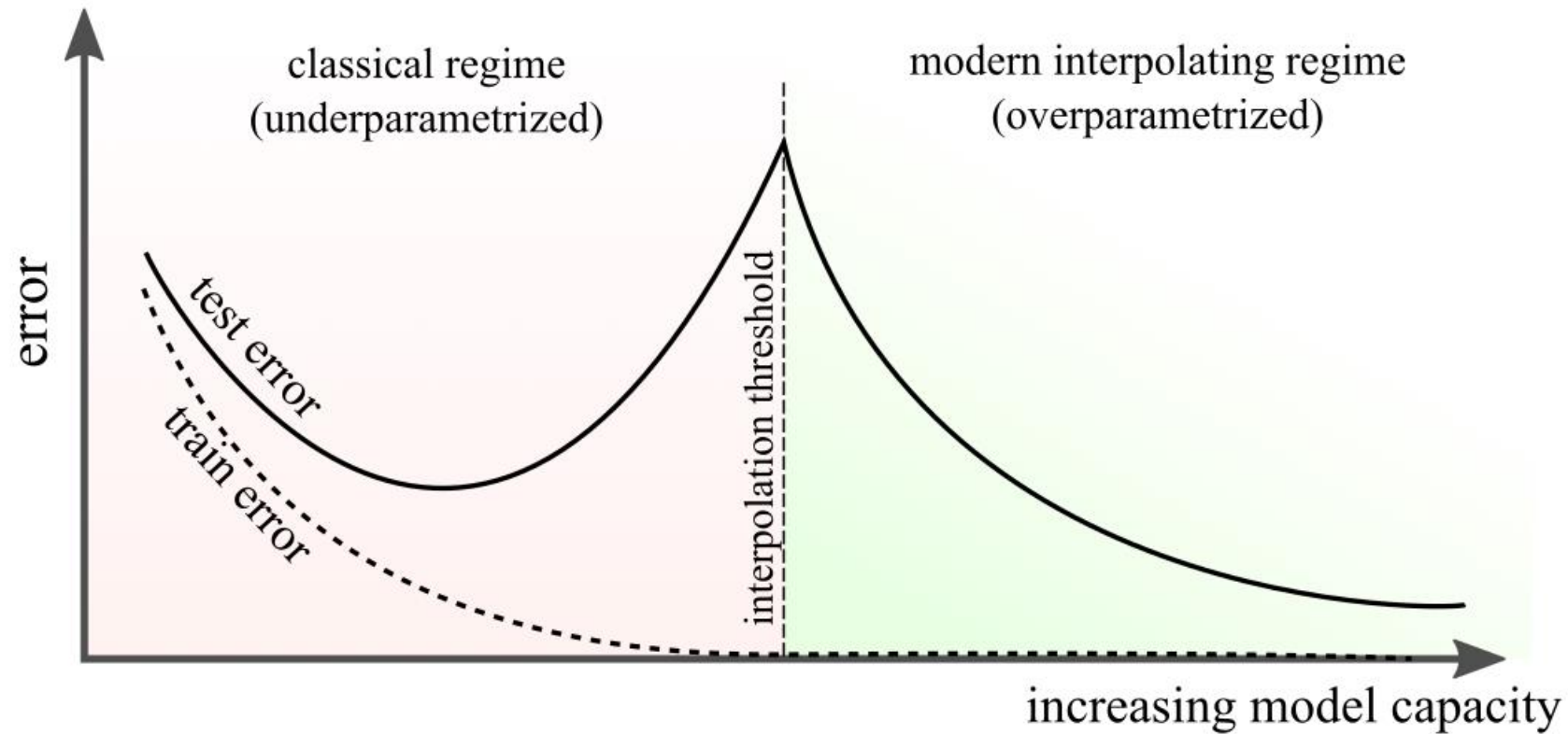
Model Selection for Overparameterized Models

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Statistics Program, KAUST

Double Descent and Neural Scaling Laws



- **Double descent** challenged common wisdom on overfitting!



Double Descent and Neural Scaling Laws



- **Neural scaling laws** allow one to extrapolate generalization wrt to model size, compute, and data set size.

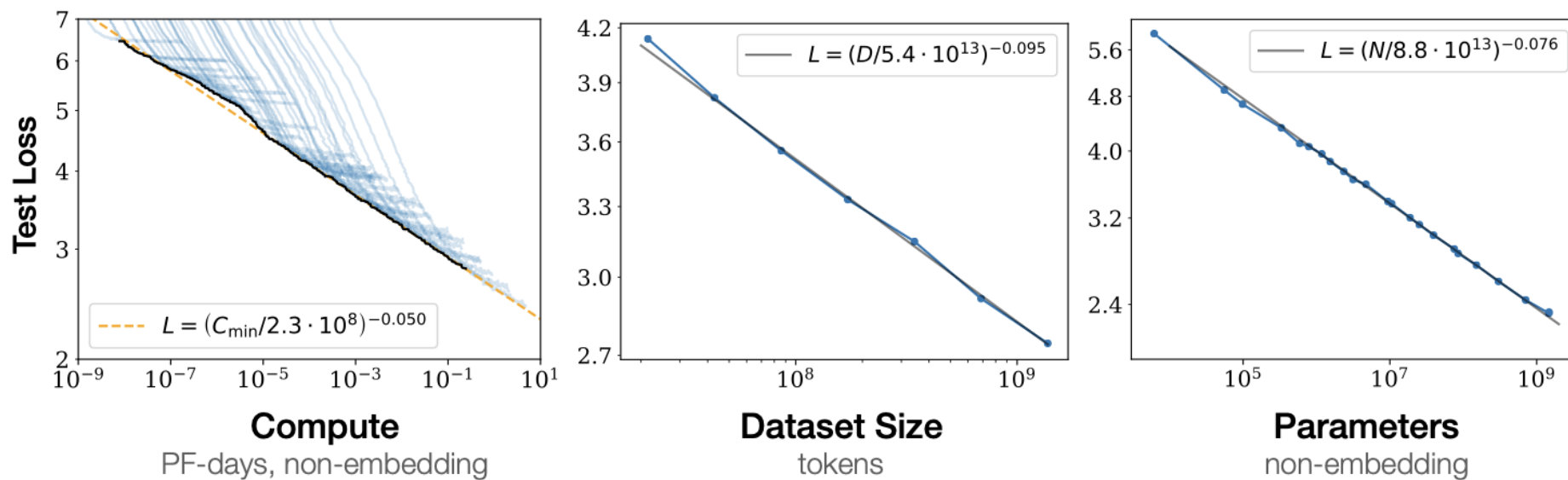


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute² used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

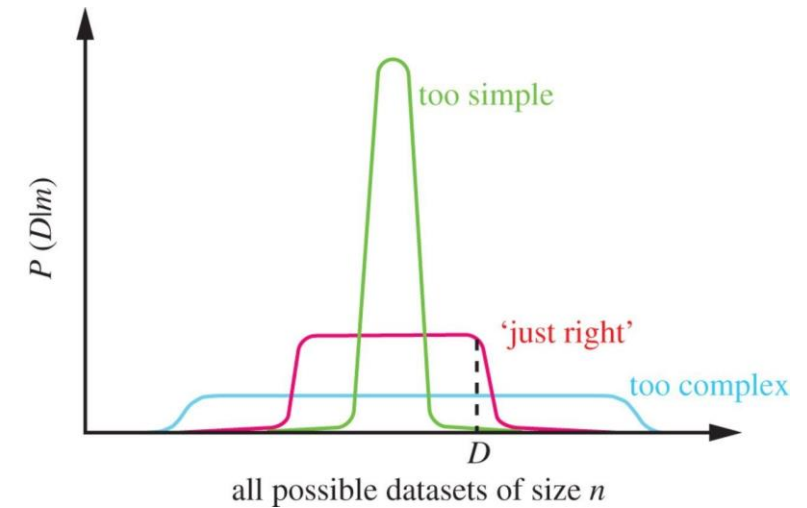
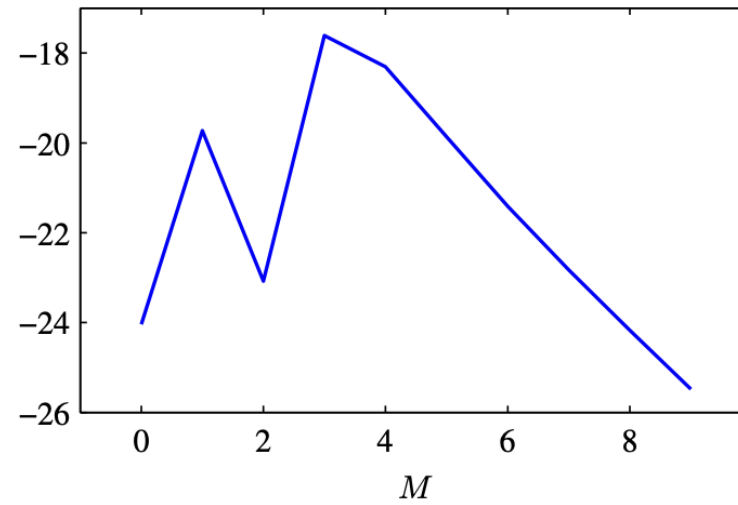
Bayesian Model Selection



- Marginal Likelihood as a model selection criterion.

$$p(\mathbf{W} | \mathbf{Y}, \mathbf{X}) = \frac{p(\mathbf{Y} | \mathbf{X}, \mathbf{W})p(\mathbf{W})}{\int p(\mathbf{Y} | \mathbf{X}, \mathbf{W})p(\mathbf{W}) d\mathbf{W}}$$

Plot of the model evidence versus the order M , for the polynomial regression model, showing that the evidence favours the model with $M = 3$.



Bayesian Model Selection

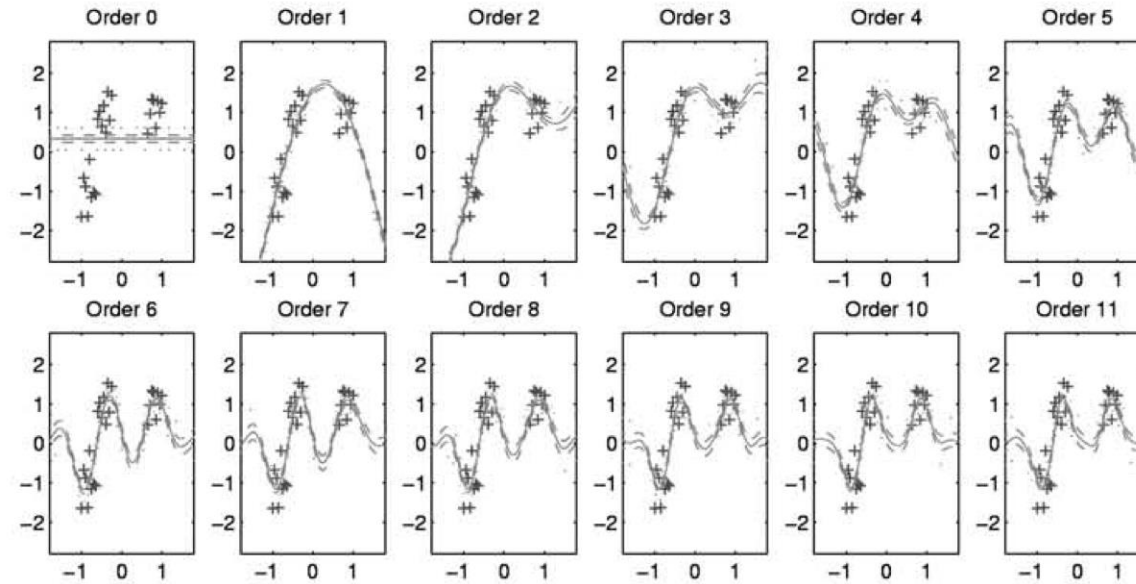


- **Bayesian Model selection** is connected with:
 - **Occam's Razor**: simplest explanation is usually the best one
 - **Kolmogorov complexity**: length of a shortest computer program generating the weights of the network
 - **Minimum Description Length**: find the most concise representation of the data
- How do we reconcile Bayesian model selection with the conclusions from research on scaling laws?

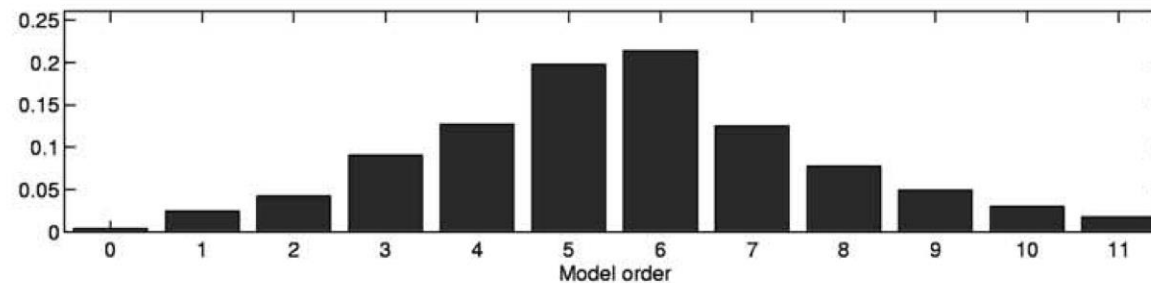
Penalizing Complexity



- Consider **trigonometric regression** as follows:
- $$y(x) = a_0 + \sum_{d=1}^D a_d \sin(dx) + b_d \cos(dx),$$



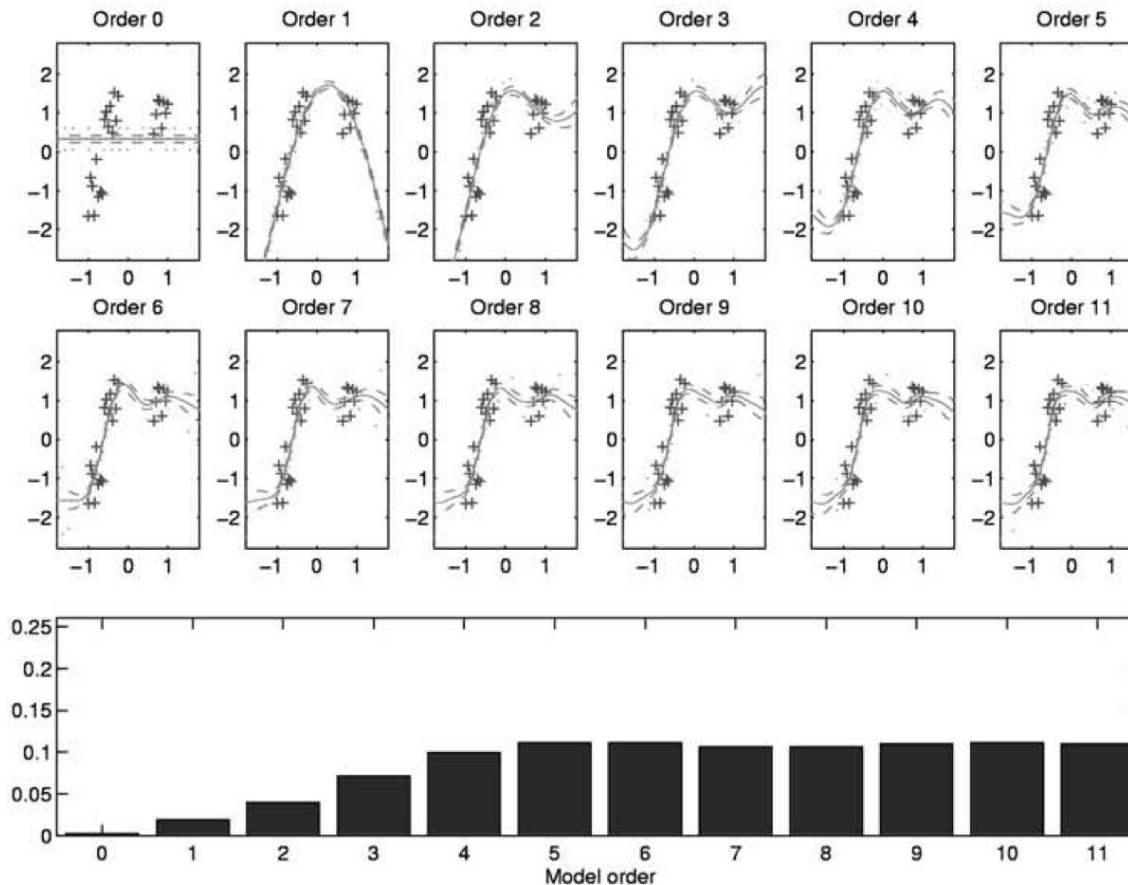
Gaussian prior with
variance independent of d



Penalizing Complexity



- Consider **trigonometric regression** as follows:
- $$y(x) = a_0 + \sum_{d=1}^D a_d \sin(dx) + b_d \cos(dx),$$



Gaussian prior with
precision proportional to d^3

Also, think about Gaussian
processes approximated with
random Fourier features
(frequencies decay fast)

Is Penalization the Answer? Or Just Over-Parameterization?



- Imposing **penalization on complexity** makes Bayesian model selection compatible with scaling laws
- However, scaling laws emerge even **without** this type of regularization*
- Is there any more to say about **priors****?
- Does this imply that in overparameterized models simplicity **emerges** over complexity***? Simplicity bias!

* except for stochastic optimization / weight decay / dropout

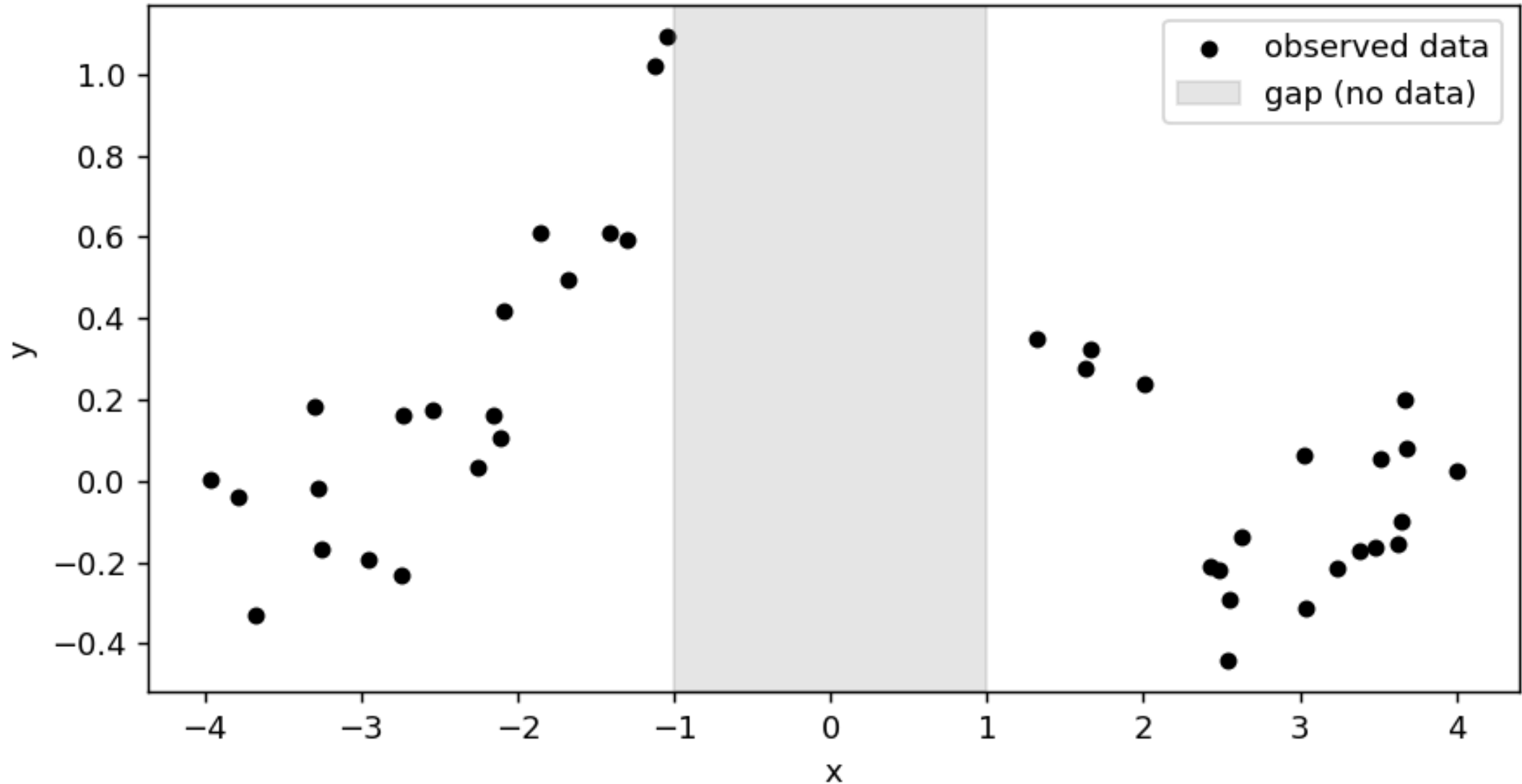
** Polson and Sokolov, arXiv 2025, Riha et al., arXiv 2026

*** Mingard et al., arXiv 2019l – Wilson, arXiv, 2025

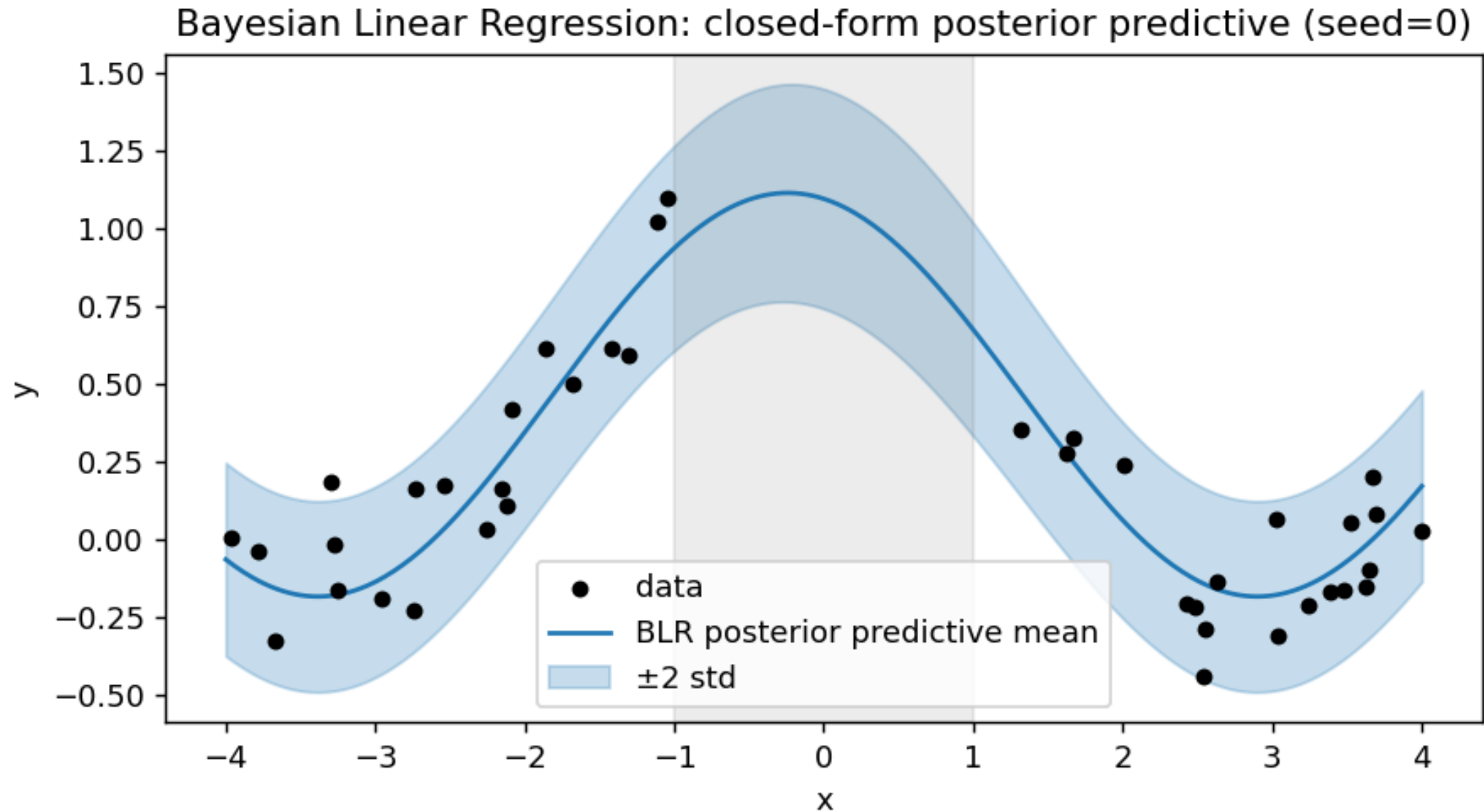
Example: Regression with Neural Networks



Dataset: simple trig-basis linear model + gap (seed=0)



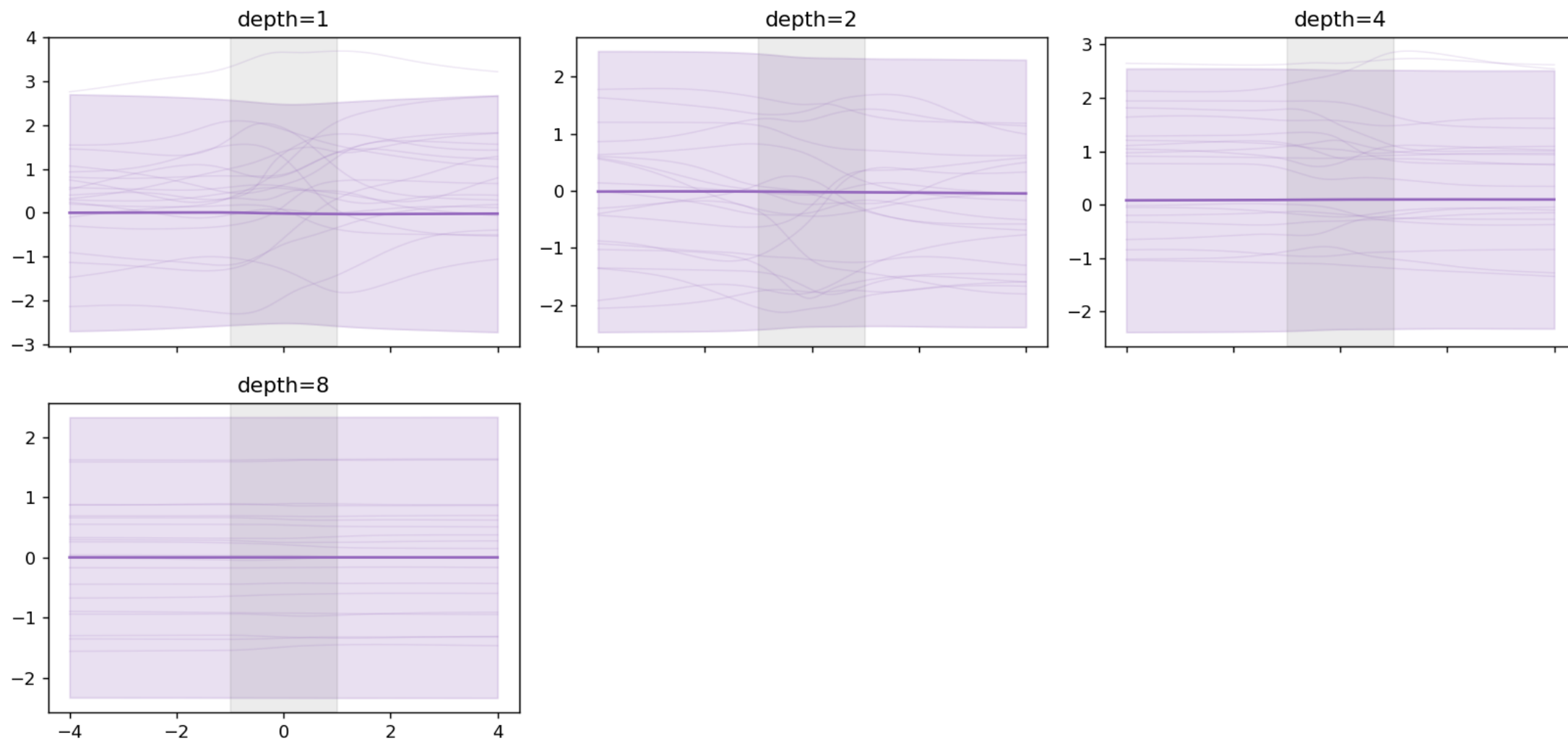
Solution with Correct Model Specification



Prior Predictive Distributions



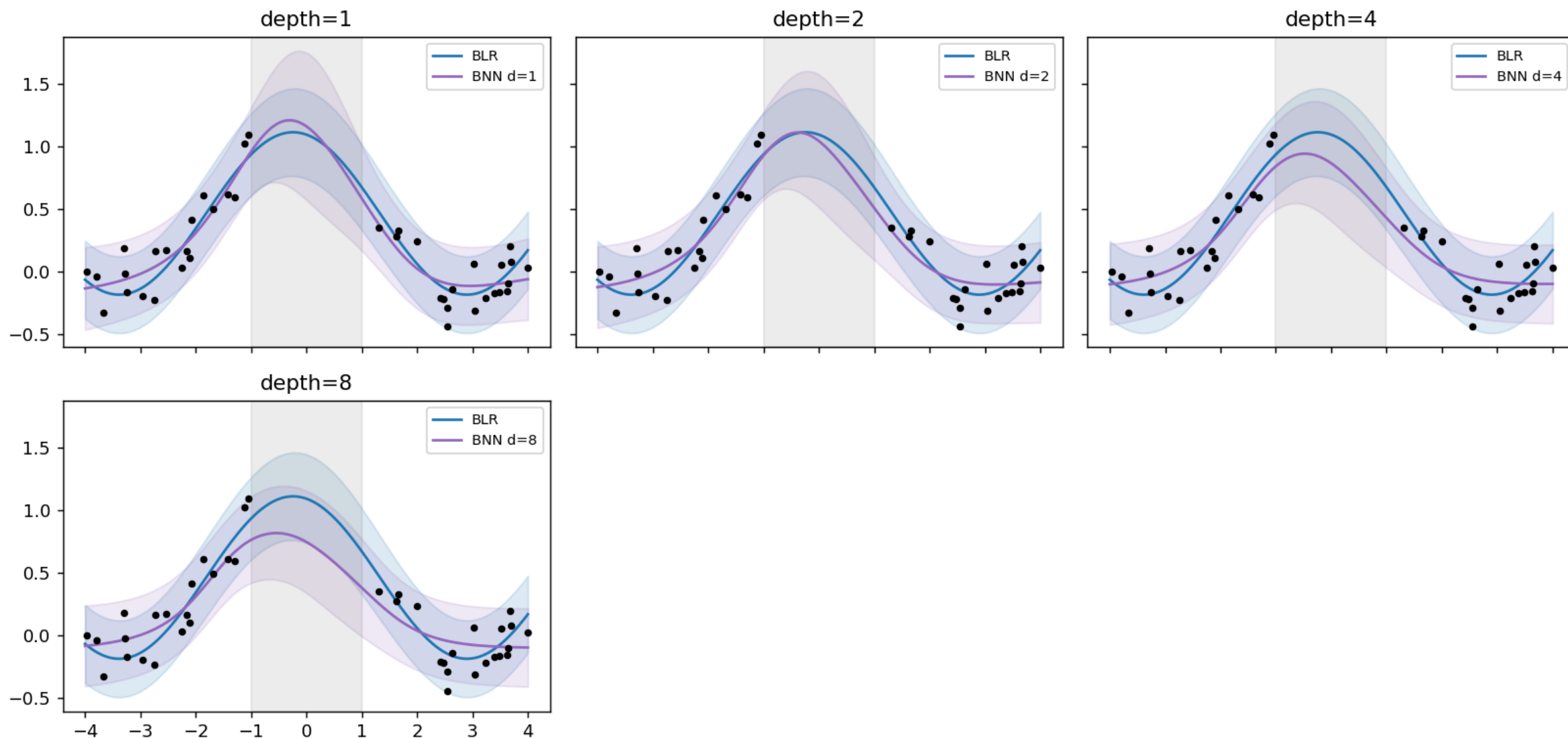
BNN prior predictive ($1/\sqrt{\text{fan_in}}$ -scaled prior, tanh activation, no data) across depths



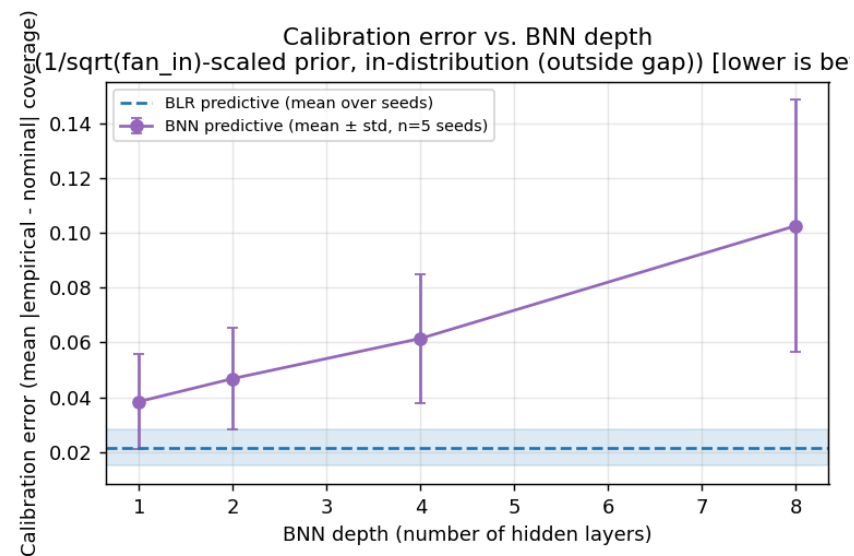
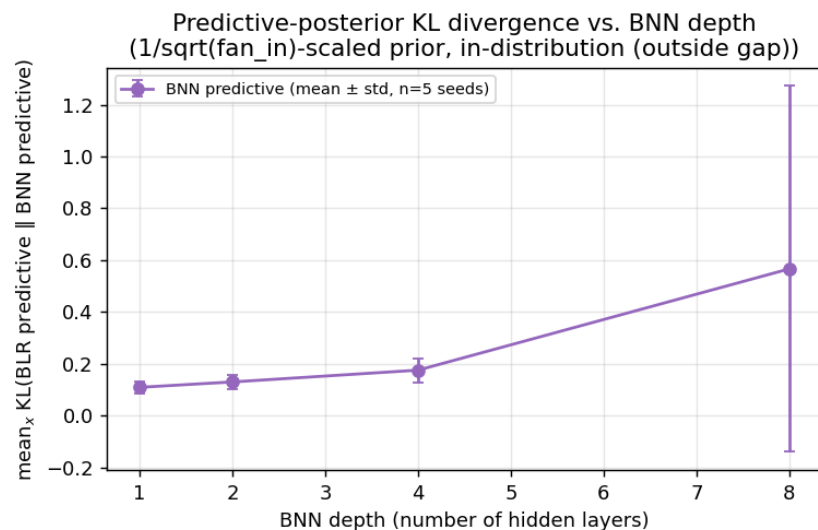
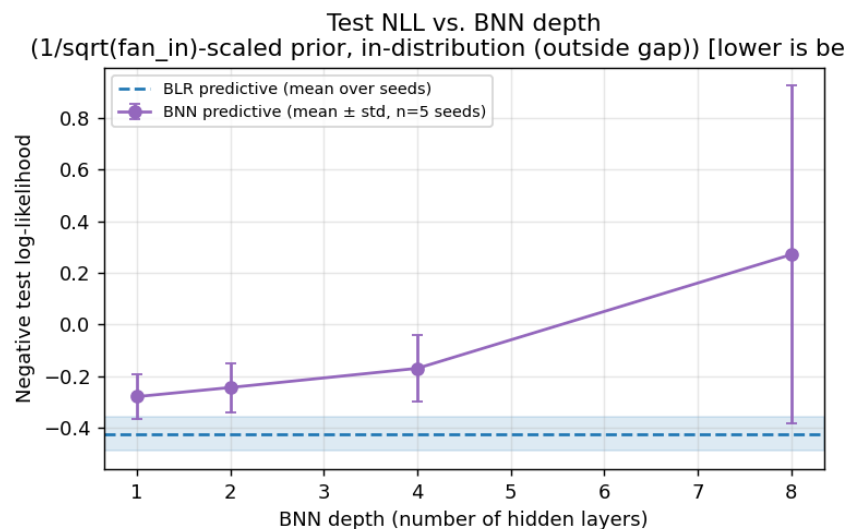
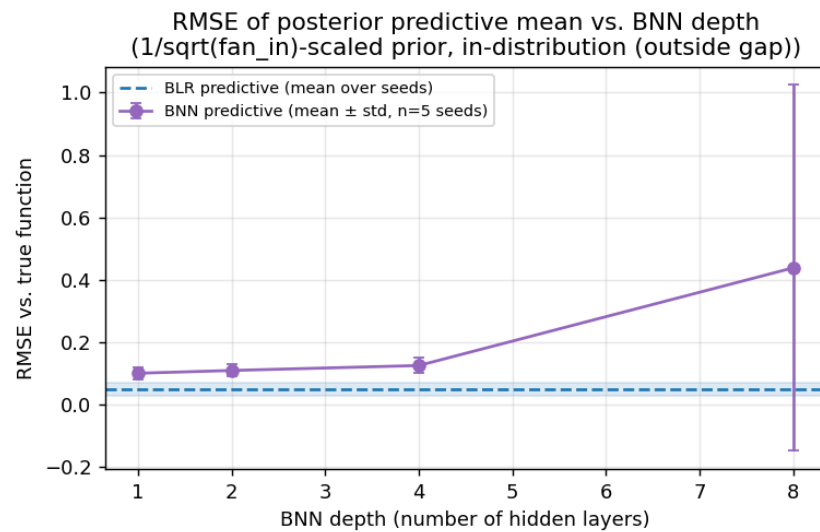
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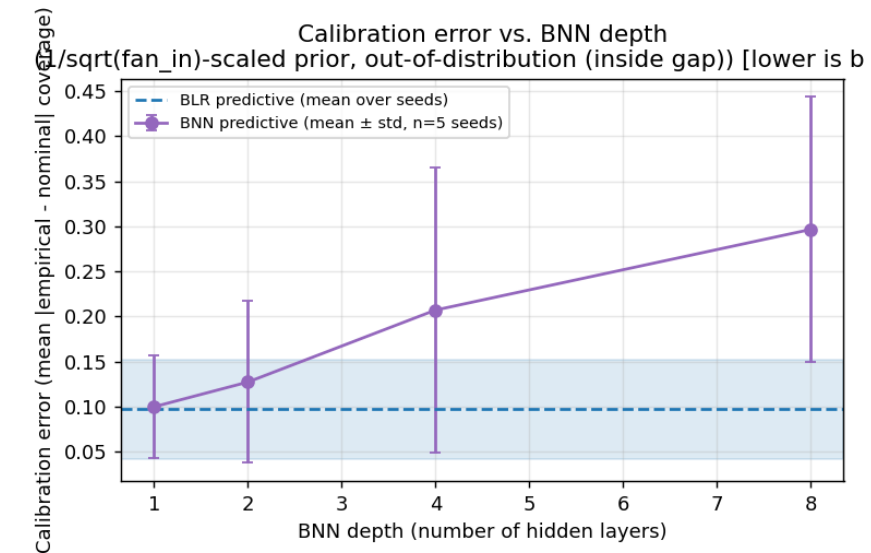
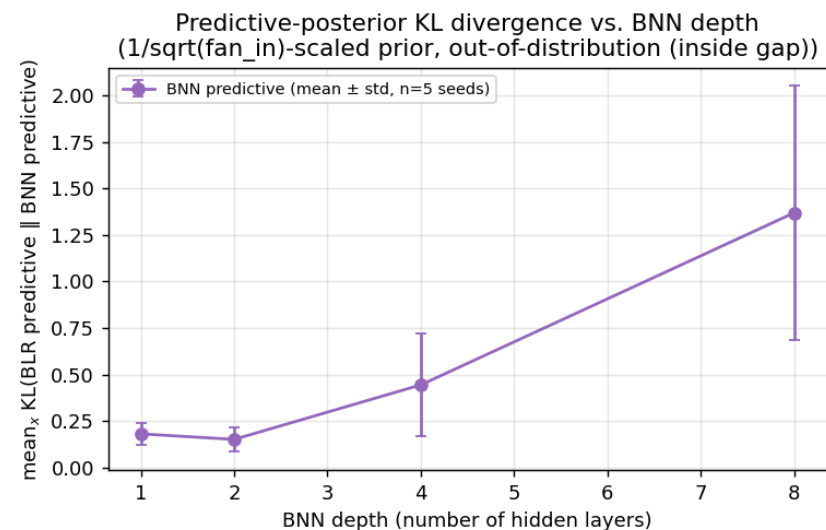
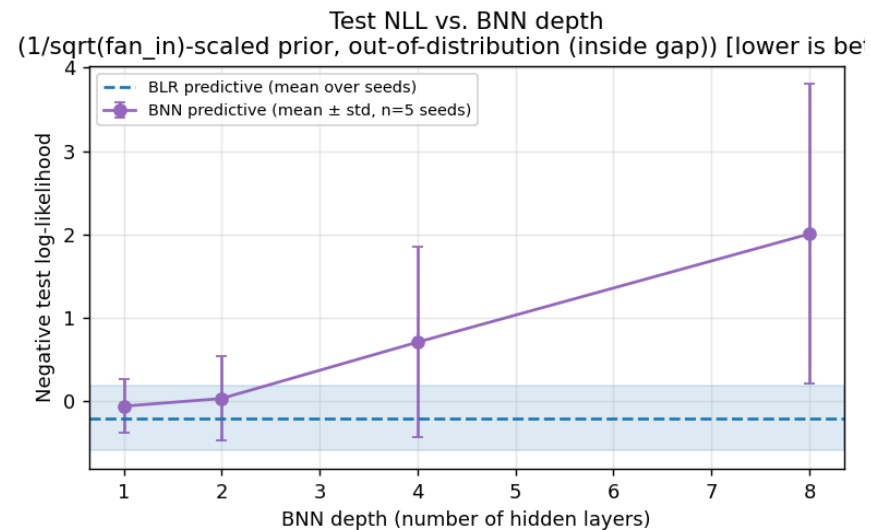
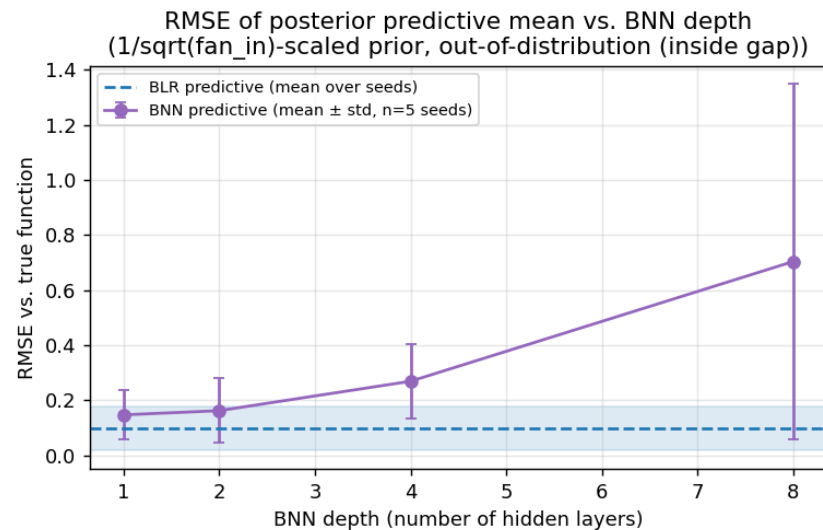
BLR (closed form) vs BNN ($1/\sqrt{\text{fan_in}}$ -scaled prior, tanh act.) posterior predictive (seed=0)



Performance In-Distribution



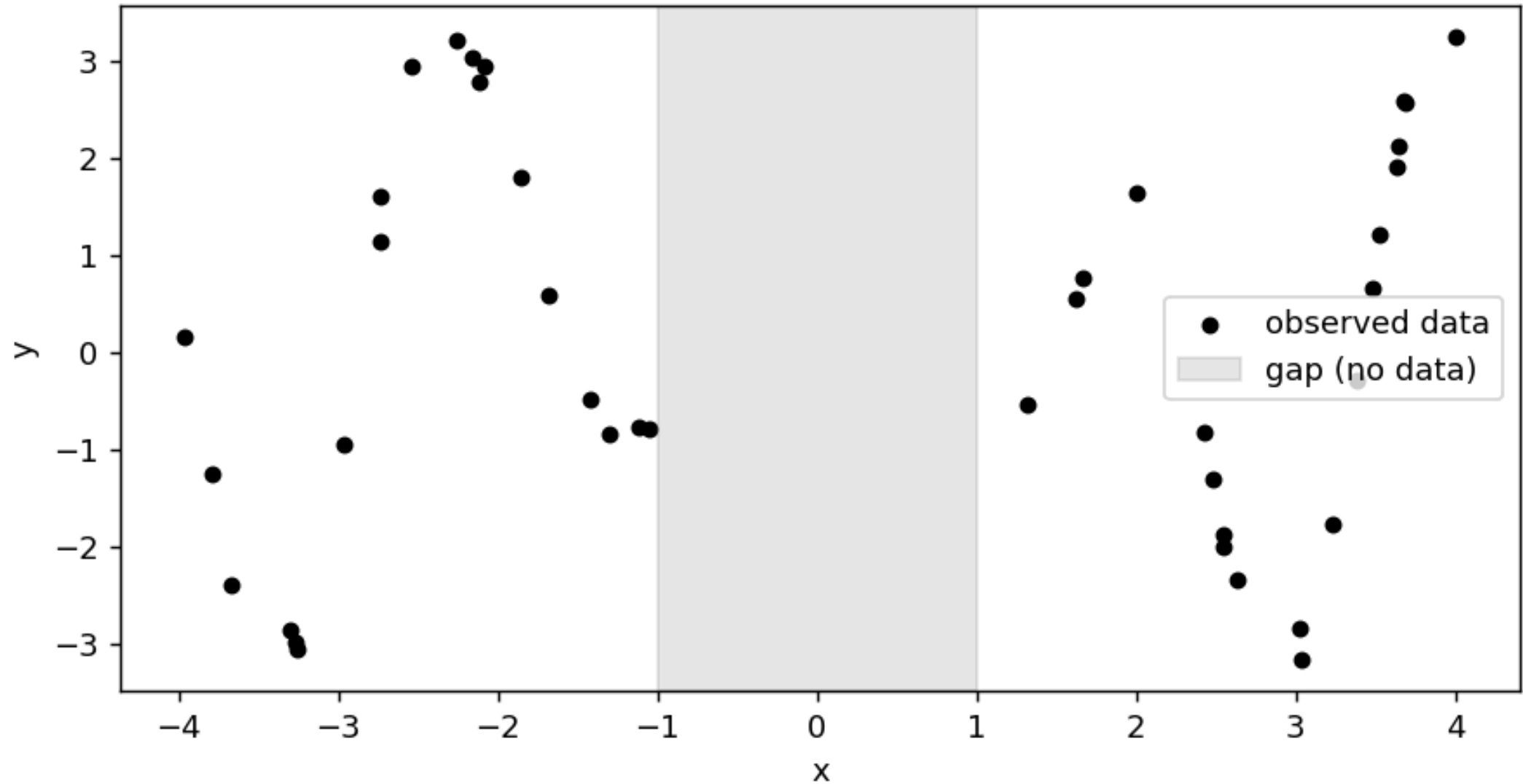
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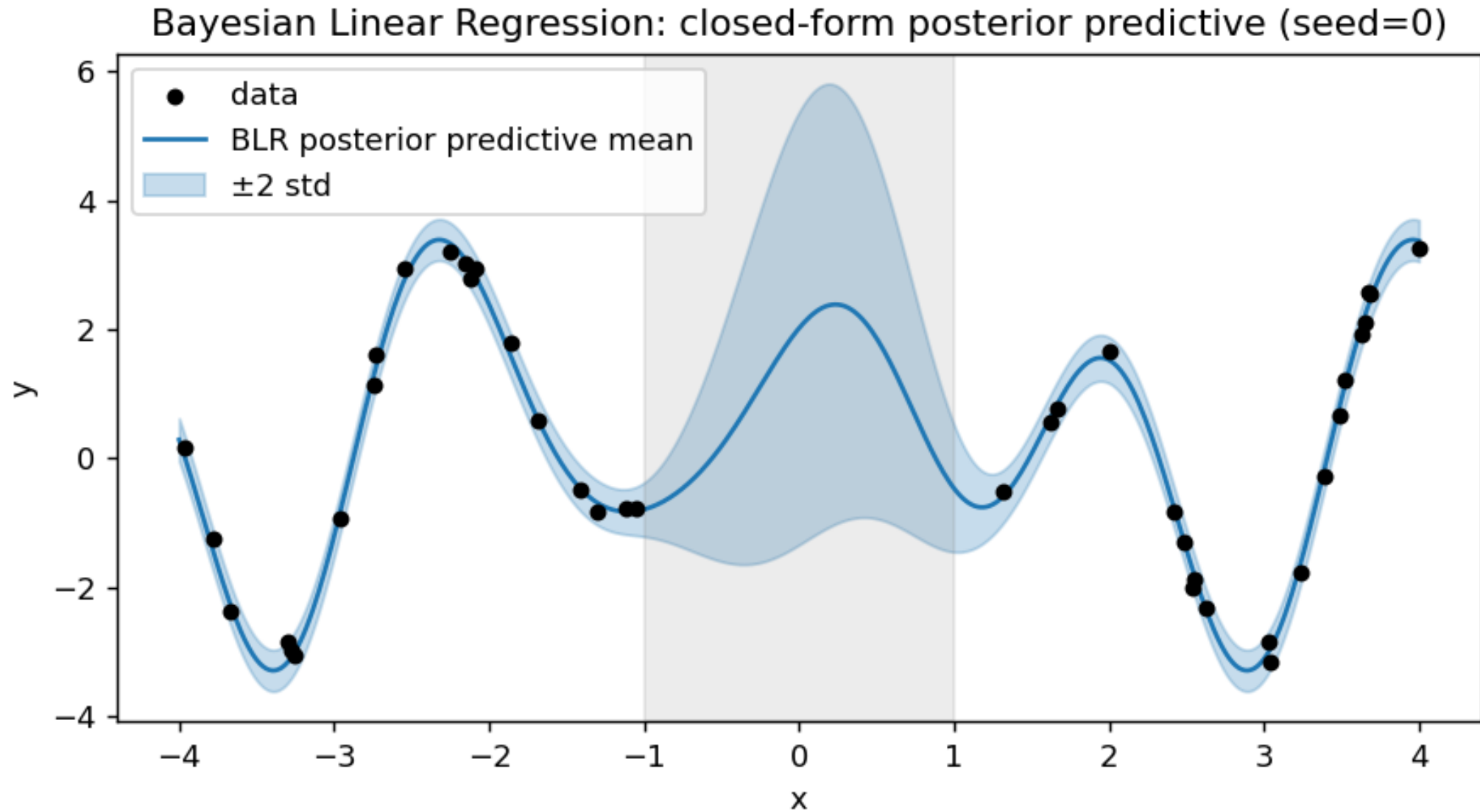
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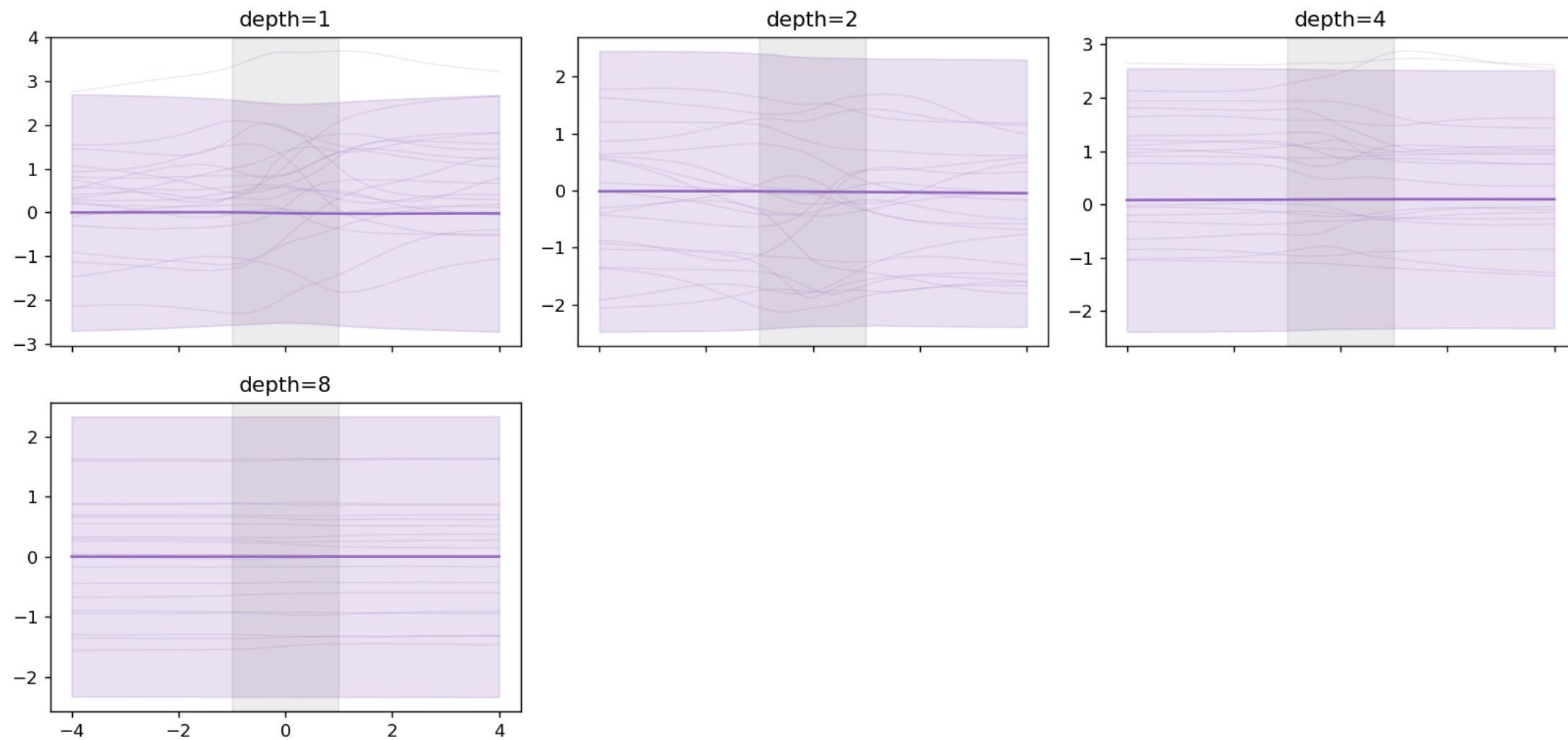
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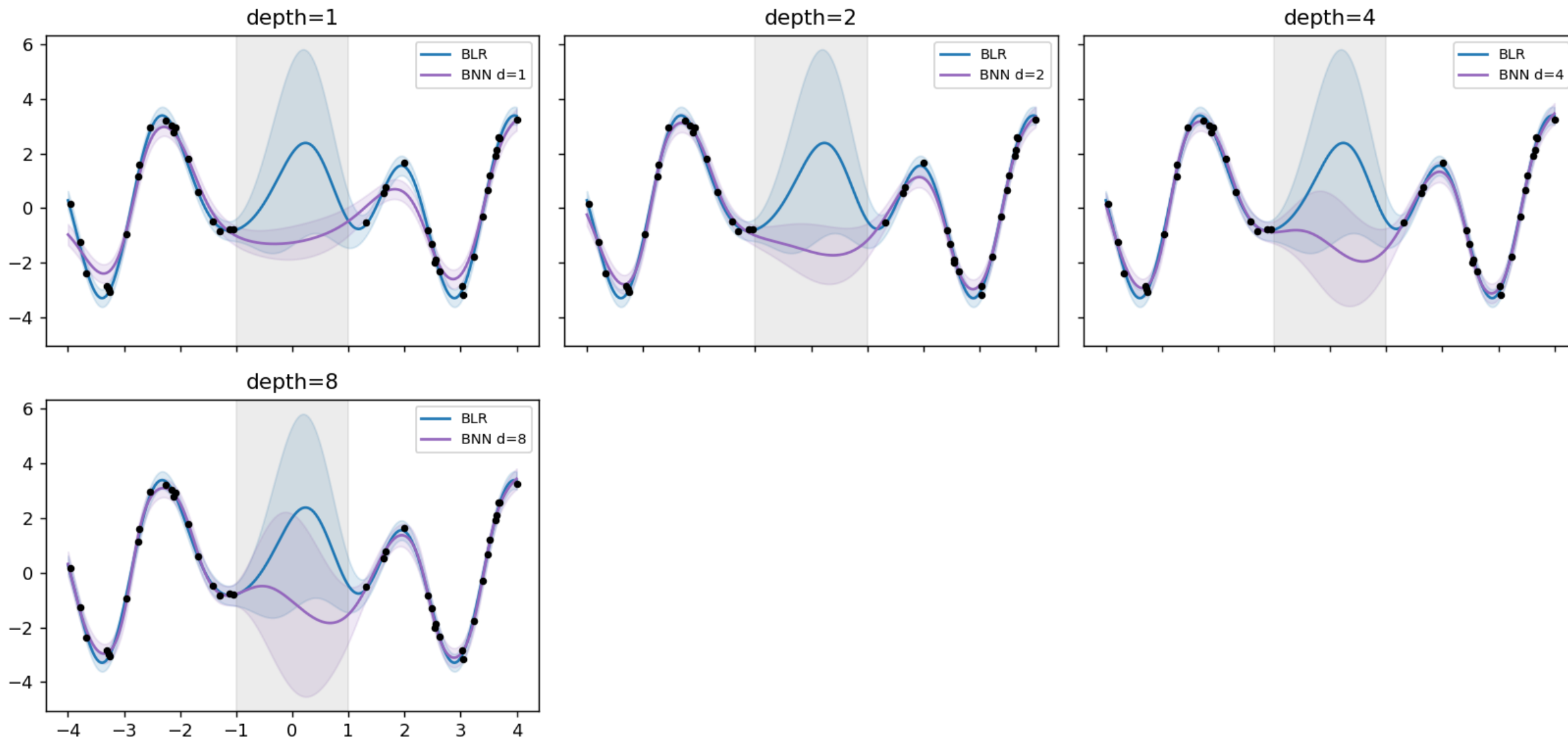
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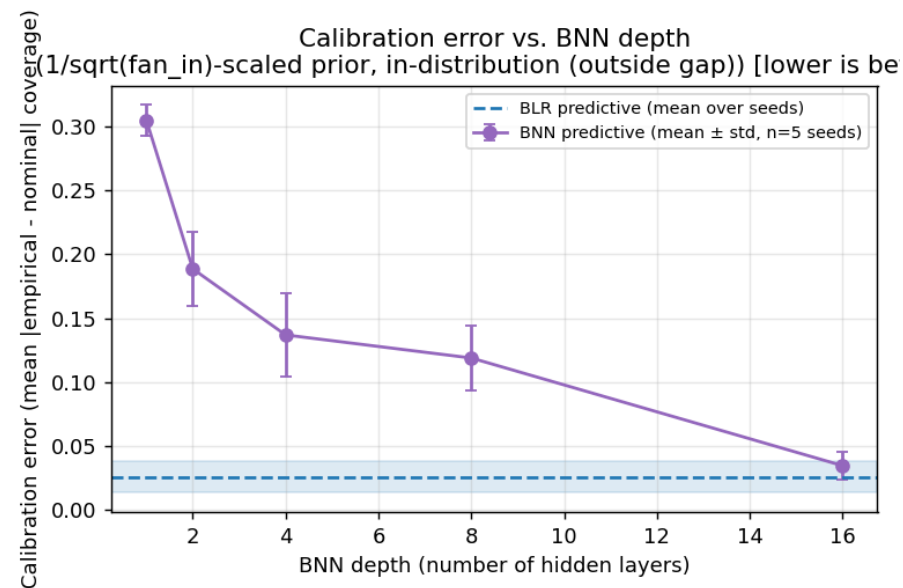
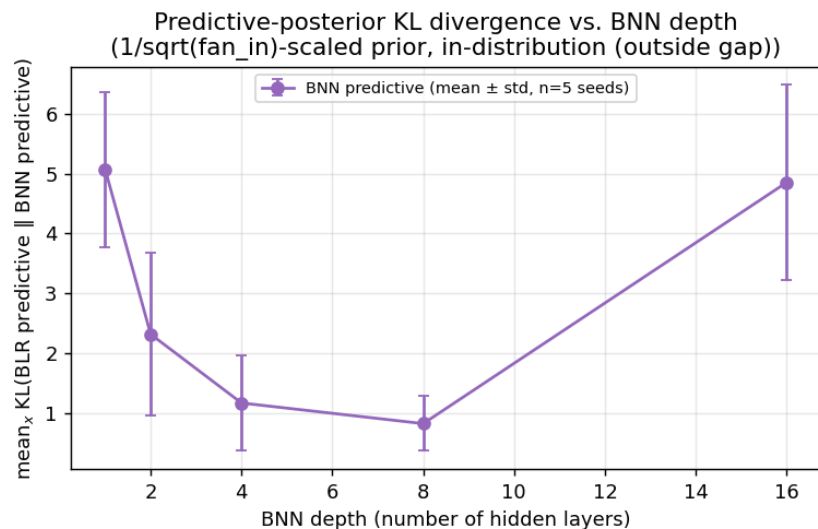
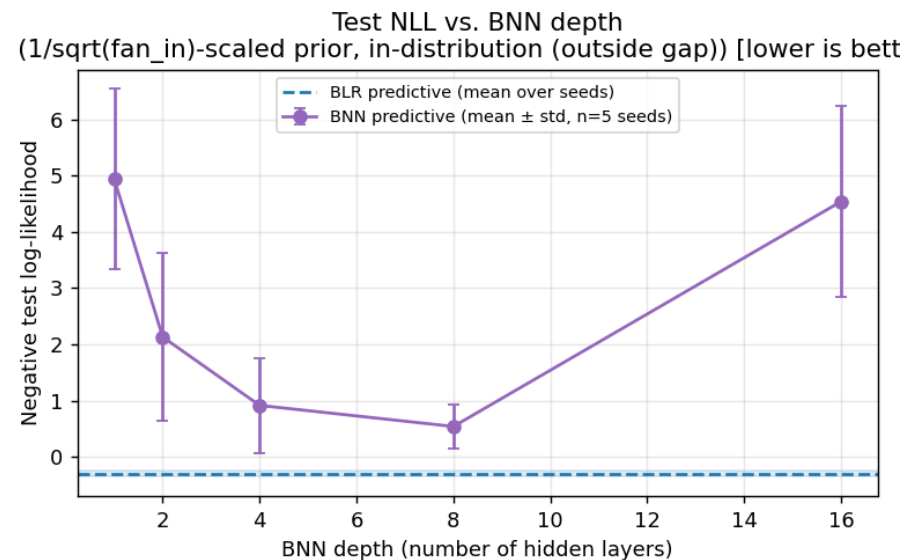
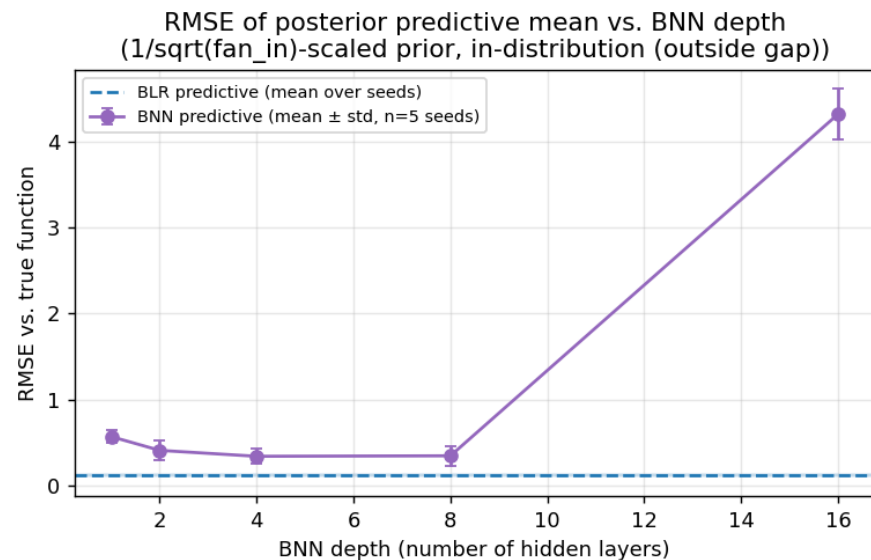
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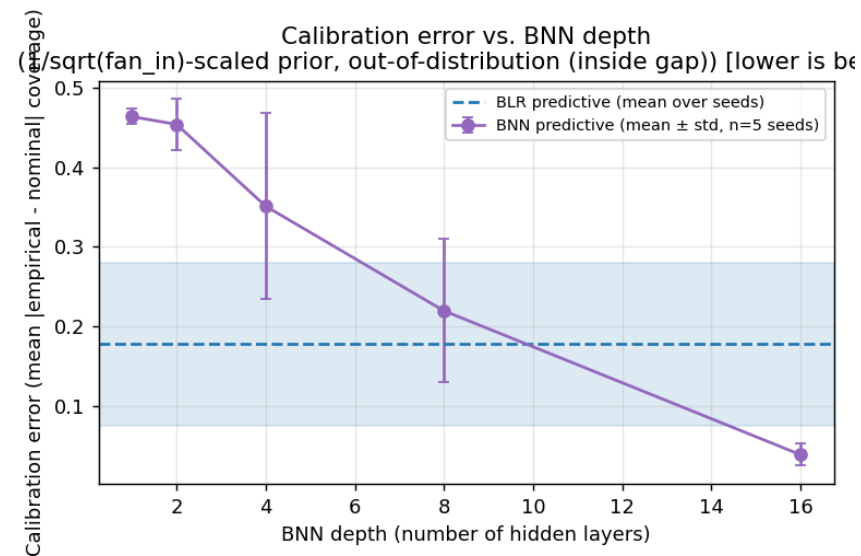
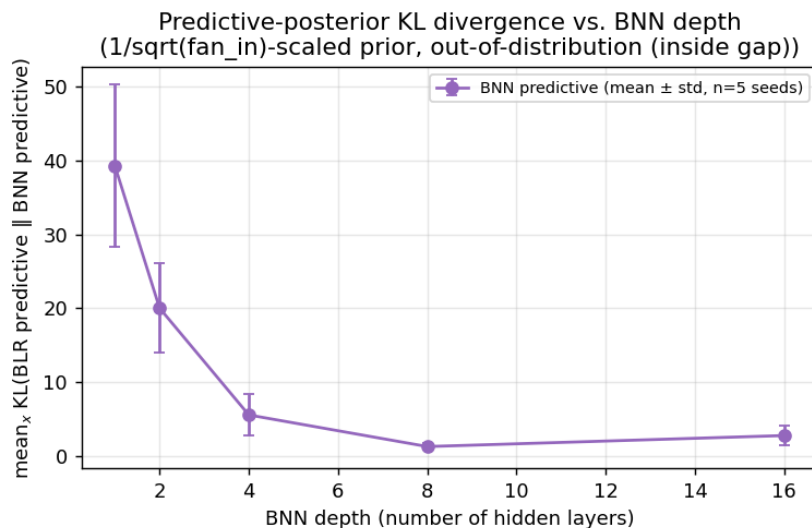
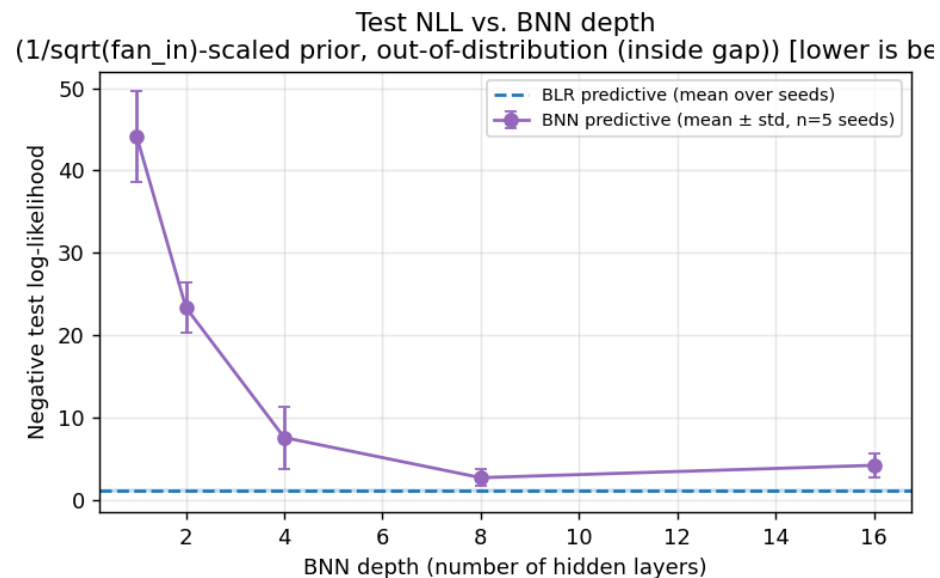
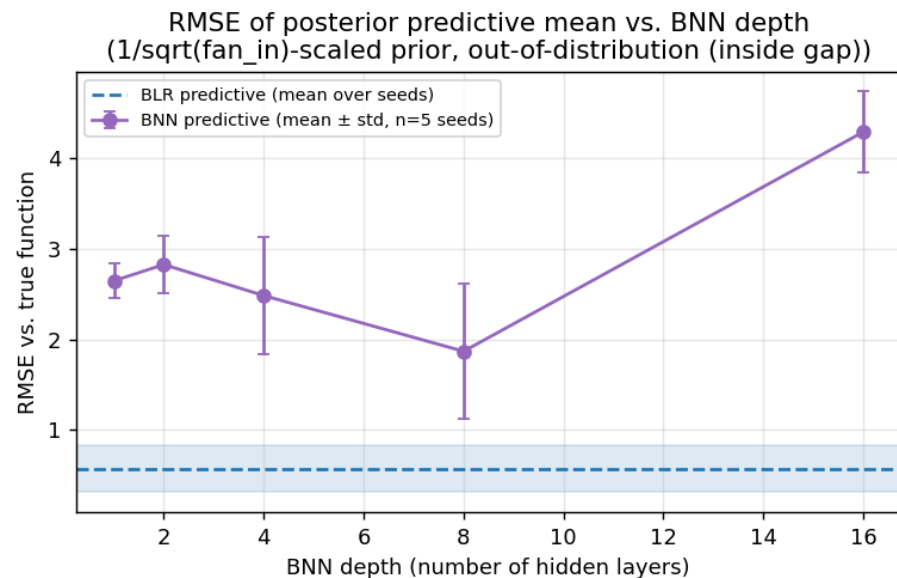
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Performance In-Distribution



Performance Out-of-Distribution



Concluding Thoughts



- Scaling Laws are a fascinating phenomenon characterizing overparameterized models
- Bayesian model selection is “compatible” with scaling laws if we introduce complexity penalties
- But what if we don’t?
- Isn’t it still relevant to look for simpler and cheaper models, as opposed to giant and expensive ones?
- How to do model selection in a practical way?